

SUNSPOKE · DESIGN PRECIS

SunSpoke design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

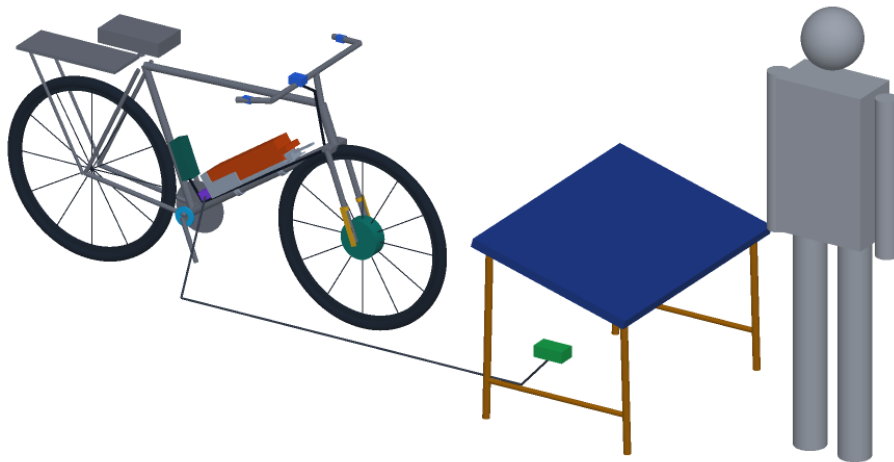
Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (SSP-DDR-001), build to SwapCell interface v0.3 (items W, C, V), numbers checked against SSP-CAL-001, parametric model and GA drawing SSP-DWG-001
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, fork torque, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunSpoke design precis

SunSpoke converts a steel roadster bicycle into a pedal-assist e-bike with a 48 V, 250 W geared front hub motor, a sealed controller and a SwapCell pack clamped inside the main triangle, all bolted on with no welding, and charges the pack from one 100 W solar panel or a shared village hub. The sizing note SSP-CAL-001 gives about 37 km of loaded range on one pack, about 315 Wh stored per 4.5 sun-hour day from one panel (about 27 km of loaded riding), 6.65 kg of added mass, and a bike kit costing about \$188 in parts, with the solar set (about \$86) costed separately and the pack priced in SwapCell. No requirement is shown to be not met; motor heating on a long loaded climb (R3, now handled by a thermistor derate rather than a cutout), fork slot fit (R4) and stopping distance (R11) are at risk.

SunSpoke

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. SunSpoke fitted to a 28 in roadster, with the 100 W solar charging set and a 1.75 m person for scale. Kit parts are colored; the donor bicycle is grey. Rendered from the parametric model `cad/src/model.py`.

■ How it works

1. **Charge.** A 100 W panel on a simple stand feeds a boost MPPT charger that raises the panel's roughly 18 V to the pack's charge voltage. A 5 m cable reaches the charge inlet on the host adapter, so the panel sits in sun while the bike and pack stay in shade. Alternatively the rider swaps the pack at a village hub that has a SwapCell dock.

2. **Store.** A SwapCell pack (13S2P, 46.8 V nominal, about 468 Wh) slides connector first down the receiver cradle on the down tube, and an over-centre lever preloads it against its end stop (SwapCell latch class V1).
3. **Wake and enable.** The cradle receptacle carries the 10 k Ω INTERLOCK coding resistor of SwapCell interface v0.3, in series with the handlebar power switch. Switching on closes the loop and wakes the pack with no supply from the bike. The pack enables legacy discharge after 2 s, which powers the host adapter; the adapter then sends the CAN heartbeat (mode 2). When the charger is connected the adapter, powered from the charge inlet, requests mode 3, or mode 4 (charge-discharge) if the bike is switched on.
4. **Sense.** A split-disc pedal-assist sensor at the crank detects pedaling. Magnetic sensors on both brake levers detect braking.
5. **Control.** A sealed 48 V, 15 A sine-wave controller on the seat tube drives the motor only while the rider pedals and no brake is applied, up to 20 km/h by default. It reads a thermistor in the motor winding and reduces current progressively from 110 °C so the winding never passes 120 °C; it never cuts assist abruptly on a climb. A handlebar unit sets the assist level, shows pack charge and has a 6 km/h walk-assist button. There is no throttle.
6. **Drive.** A 250 W geared hub motor, laced by a local mechanic into a standard 28 in (ETRTO 635) rim, replaces the front wheel. Torque arms on both fork blades carry the motor's reaction torque so it does not act on the thin dropouts alone.

Front drive leaves the roadster's chain, freewheel or coaster brake and rear carrier untouched, which is what makes a bolt-on fit and local repair realistic. If the host adapter fails, the pack still runs the bike in SwapCell legacy mode (discharge only, 15 A, the same as the controller limit), so the rider gets home.

SunSpoke: daily solar energy flow, Wh per day (estimates, 4.5 peak sun hours)

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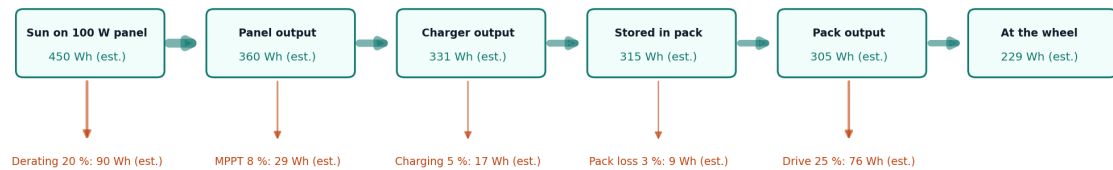


Figure 2. Daily solar energy flow from one 100 W panel at 4.5 peak sun hours, in Wh per day. All values are estimates from SSP-CAL-001.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`. The general arrangement is drawing SSP-DWG-001 (`cad/drawings/SSP-DWG-001.pdf`), generated from `cad/src/model.py`.

#	Component	Choice	Notes
1	Front hub motor wheel	48 V, 250 W geared hub, slow winding, 100 mm spacing, 10 mm axle flats, built-in 10 k Ω NTC winding thermistor, laced into a 28 in steel rim	Donor tyre and tube reused. Decided by Amish, 2026-09-25
2	Torque arms	5 x 20 mm steel plate arms keyed to the axle flats, clamped 130 mm up each fork blade	Mandatory on both sides
3	Controller	48 V, 15 A limit, sine-wave, potted or IP65, pedal-assist, brake cut-off, power-lock, walk-assist and motor thermistor inputs; derate from 110 °C	Generic part; thermistor input decided by Amish, 2026-09-25 (SSP-DDR-002)
4	SwapCell receiver cradle	Folded steel cradle with guides, latch catch, receptacle with 10 k Ω coding resistor, over-centre preload lever; two lined band clamps	Latch class V1 vehicle receiver; clamps to 28 to 32 mm down tubes; no welding
5	Host adapter and charge port	Microcontroller with CAN transceiver, diode-OR supply from PACK+ and the charge inlet, controller power-lock output	SwapCell vehicle host for modes 2, 3 and 4
6	SwapCell pack	13S2P, about 468 Wh, 340 x 90 x 80 mm, about 2.85 kg	Priced once in SwapCell (\$414); not in the kit cost
7	Pedal-assist sensor	Split 12-magnet disc and Hall sensor	Fits without pulling the crank
8	Handlebar control and brake cut-off	LED level and charge display, power switch in the INTERLOCK loop, walk-assist button, two clamp-on lever sensors	Works with rod or cable brake levers
9	Wiring harness	Keyed waterproof connectors, 20 A fuse, spiral wrap	Every joint pluggable (R9)
10	Solar panel	100 W monocrystalline, about 1,000 x 670 mm	Costed separately; can be shared at a hub
11	Boost MPPT charger	15 to 25 V in, CC/CV to 54.6 V at up to 2 A out	Generic; the host adapter is the SwapCell charge host
12	Panel stand	Local steel angle or timber A-frame, 10 to 20 degree tilt	Local make
13	Charge cable	5 m outdoor cable with keyed plug to the charge inlet	Pack charges in shade

SunSpoke: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Front hub motor, 250 W geared
- 2 Torque arms (pair)
- 3 Controller, sealed
- 4 SwapCell receiver cradle, V1 lever
- 5 Host adapter (CAN, charge port)
- 6 SwapCell pack (not in kit cost)
- 7 Pedal-assist sensor
- 8 Handlebar control and brake cut-off
- 9 Wiring harness with fuse
- 10 Solar panel, 100 W
- 11 Boost MPPT charger
- 12 Panel stand (local make)
- 13 Charge cable

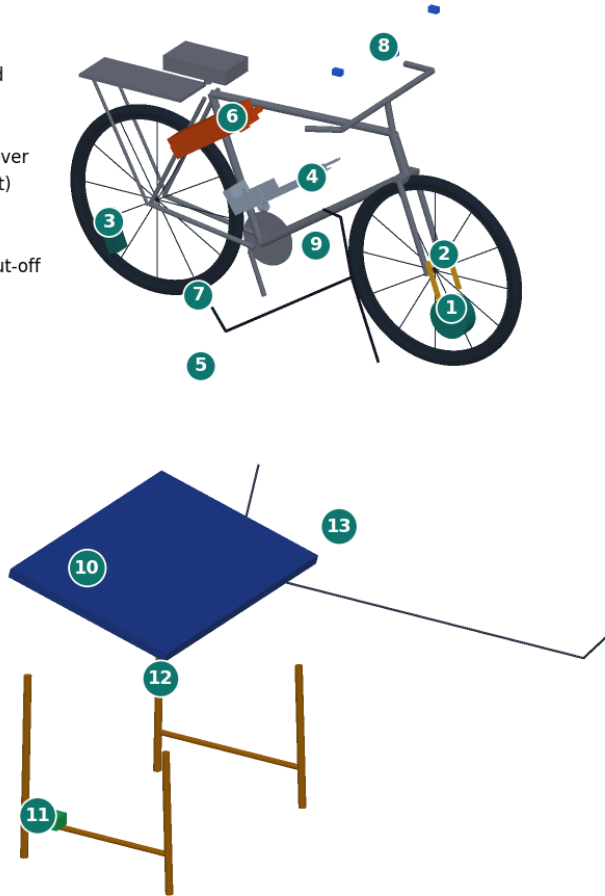


Figure 3. Exploded view with numbered callouts matching the BOM. The solar charging set is shown displaced below the bike.

■ Key numbers

All values come from SSP-CAL-001 (docs/04-calcs/sizing.py) and are paper estimates. The design load case is 128.65 kg (75 kg rider, 25 kg cargo, 22 kg roadster, 6.65 kg of kit and pack) on a dry dirt road.

Quantity	Value	Basis	Requirement
Pack energy, design case	about 11.3 Wh/km	Crr 0.02, CdA 0.6 m ² , 18 km/h, 30 % allowance, rider 70 W, drive 75 %	
Range on SwapCell	about 37 km design (36 with minimum cells); 75 km light; 23 km with 80 kg cargo	419 Wh usable	R2 met
Motor on the 8 % grade at 8 km/h	204 W and 32.5 N·m (81 % of a 40 N·m peak)	127.6 N, rider 80 W	R3 met on power
Motor winding, top of 500 m at 35 °C	about 100 °C base case; about 139 °C hot case	Lumped thermal model	R3 at risk
Thermal derate from 110 °C	Starts after 697 m (base) or 225 m (hot) of the 8 % climb; sustained 4.7 or 3.7 km/h on an unending climb	Winding held at 120 °C, rider 80 W	R3
Added mass	6.65 kg (kit 3.80, pack 2.85)	Table 2 of SSP-CAL-001	R6 met, 0.35 kg margin
Solar, one 100 W panel	315 Wh/day stored (280 at 4 h); about 27 km of loaded riding per day	20 % derating, 92 % MPPT, 95 % charging	R8 met
Recharge 10 to 100 %	1.3 days (1.5 at 4 h)	421 Wh / 315 Wh per day	R8 met
Peak charge current	1.47 A (0.15C); 1.39 A net in mode 4	74 W into about 50 V	R13
INTERLOCK node	0.30 V, 30 µA	10 kΩ against a 100 kΩ pull-up	R13
Torque arms	154 N per clamp; 60 MPa in the arm, safety factor 4.2	40 N·m peak at 130 mm	R10
Dropout slot filing	about 0.24 mm per side on a 9.53 mm slot	10 mm flats	R4 at risk
Cradle retention at 25 g	Axial margin 6.8; rotation margin 1.40	3.5 kg pack plus 1.0 kg cradle	R13, not verifiable
Pack fit in the triangle	123 mm clearance; 164 mm free travel for removal (143 needed)	Parametric model, pack at 48 % of the down tube	R5 met on paper
Stopping from 20 km/h	8.5 m dry; 21.7 m wet	Rod brakes on steel rims, 0.5 s delay	R11 at risk
Bike kit cost	about \$188	Items 1 to 5, 7 to 9, 14	R12 met
Solar set cost	about \$86	Items 10 to 13, costed separately	
SwapCell pack	\$414, priced in SwapCell (SWC-CAL-001)	Excluded from the kit	

Fork and frame torque

The motor pushes against its own axle. At the assumed 40 N·m peak, the axle flats would load each bare dropout face with about 4 kN, enough to spread or crack thin stamped dropouts over time. Torque arms move the reaction to clamps 130 mm up each blade, about 154 N each. Roadster slots are often 3/8 in (9.53 mm), so the 10 mm flats

may need about 0.24 mm filed from each face; whether that is acceptable with torque arms fitted is open (SSP-DDR-001 item 15). Braking and pothole loads also rise with the heavier front wheel and higher speed; the fork's fatigue margin is unverified.

■ Key design choices

All of these were decided by Amish on 2026-09-25 (go with recommendation), in SSP-DDR-001 or SSP-DDR-002.

- **48 V on the SwapCell pack (Option A).** Meets R2, shares packs and docks with other portfolio vehicles and keeps the pitch. A 36 V kit with its own generic pack (Option B) is kept only as a later low-cost variant.
- **Budget covers the bike kit.** `budget_usd` stays at \$250 and R12 applies to the bike kit (about \$188); the solar set is costed separately and the pack is priced once in SwapCell.
- **Front geared hub motor.** Leaves the drivetrain, rear brake and carrier untouched and is the easiest bolt-on.
- **Pack in the main triangle on the down tube.** Keeps the rear carrier free and the mass low and central.
- **Motor laced locally into a 28 in rim,** with a lacing card.
- **Assist cut-off 20 km/h by default,** adjustable only by the mechanic and never above 25 km/h. SSP-CAL-001 shows the loaded bike stops in about 8.5 m from 20 km/h but about 12.4 m from 25 km/h on dry dirt.
- **Pedal assist only, plus a 6 km/h walk assist.** No throttle.
- **Charge on the bike.** The charger plugs into the host adapter's charge inlet and the adapter acts as the SwapCell charge host; hub docks remain an option.
- **SwapCell interface v0.3 items W, C and V.** The cradle fits the 10 kΩ coding resistor (W), the adapter uses mode 4 when the bike is on while charging (C), and the cradle is a latch class V1 receiver (V).
- **Power switch in the INTERLOCK loop, adapter powered from legacy discharge or the charge inlet.** Decided by Amish, 2026-09-25: go with recommendation (SSP-DDR-002). Switching off opens the pack output, a hard off that needs no electronics on the bike.
- **Controller with a motor thermistor input (R3).** Decided by Amish, 2026-09-25: go with recommendation (SSP-DDR-002). The motor carries a winding thermistor and the controller derates from 110 °C instead of cutting out.

■ Safety

Safety: SunSpoke carries a lithium-ion pack of about 468 Wh, drives a loaded bicycle faster than its rider alone, and puts new loads on an old steel fork. Treat each of these as a hazard at every stage.

- **Lithium pack.** A cell in thermal runaway vents flammable, toxic gas and can ignite its neighbors. Use only a SwapCell pack with its BMS, fuse the harness (item 9), charge on a non-combustible surface in shade and away from sleeping areas, never charge a pack that is damaged, swollen or has been submerged, and keep the pack out of direct sun when parked. The pack refuses charge below 0 °C and above 45 °C cell temperature.
- **Pack retention.** A pack that leaves its cradle at speed is a 2.85 kg projectile with live contacts. The cradle must meet SwapCell latch class V1, the lever must have its detent, and the band clamps must be tightened to their stated torque and checked at every service; rotation resistance at a 25 g lateral shock has only about 1.4 margin on paper.
- **Fork dropout failure.** A spun axle can rip the motor cable and let the wheel leave the fork, which throws the rider over the bars. Torque arms on both sides are mandatory, axle nuts need a set torque and a check at every

service, and a fork with cracked, bent or heavily filed dropouts must not be converted. Fork fatigue under the heavier wheel is unverified.

- **Braking with added speed and mass.** At 20 km/h the loaded bike carries about 2.0 kJ, about 2.5 times the unconverted bike at 13 km/h. Rod brakes on steel rims lose most of their grip in rain: about 22 m to stop from 20 km/h wet against 8.5 m dry. Brake cut-off sensors on both levers, the 20 km/h limit, and a brake check with new blocks at fitting are part of the kit. Braking with only the front brake on loose ground can wash out the front wheel.
- **Motor overheating.** On a long loaded climb in the heat the winding may exceed its limit (R3). A sudden cutout on a hill can stall a loaded bike, so the controller derates on the motor thermistor instead (decided, SSP-DDR-002); the bike slows to walking pace on a very long hot climb but keeps moving.
- **Wiring in rain.** The system is about 50 V DC, below the usual touch-safety threshold, but water in connectors causes corrosion, shorts and sudden loss of assist. Use keyed IP65 connectors with dielectric grease, drip loops, and routing that keeps the motor cable exit facing down. The fuse protects against a pinched cable shorting to the frame. Opening the power switch opens the pack output within 1 ms.
- **Traction.** With cargo on the rear carrier, the front wheel carries little weight and a front motor can spin on sand or wet laterite. Assist ramp-up must be soft.

■ Open questions

These remain after SSP-DDR-001. None of them is TRL 4 work to be started now; TRL 4 is on hold by Amish's instruction.

- First partner and region for co-design and fitting trials. Proposed, awaiting Amish.
- Whether light slot filing is acceptable with torque arms (R4). Needs the donor survey; awaiting Amish.
- Motor data for R3: winding resistance, thermal capacity and gear temperature limit of a named 250 W geared hub.
- Wet braking: whether R11 should add a wet target, and whether better blocks or a rim brake upgrade belong in the kit.
- The legacy-to-heartbeat transition in the SwapCell pack. Raising it with the SwapCell project is decided (SSP-DDR-002); the change belongs to that repo.
- Confirm that a low-cost boost charger with input-voltage tracking gives near-MPPT yield from one 18 V panel.

Concept media: [blueprint sheet](#), [interactive 3D model](#).